

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE330	COMPETITIVE CODING APPROACHES-TECHNIQUES	2	0	1	3	
Course Weightage	ATT: 5 CAP: 45 ETP: 50					
Course Focus	EMPLOYABILITY,SKILL DEVELOPMENT,ENTREPRENEURSHIP					

Course Outcomes :Through this course students should be able to

CO1 :: analyze the time and space complexity of algorithms

CO2 :: apply efficient algorithms for primality testing and number theory problems

CO3 :: apply recursion and backtracking techniques to solve computational problems

CO4 :: analyze and implement dynamic programming techniques for optimization problems

CO5 :: develop optimized solutions for classical dynamic programming problems

CO6 :: evaluate and implement efficient sorting algorithms for problem solving.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	PROGRAMMING PEARLS	JOE BENTLEY	PEARSON

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	CRACKING THE CODING INTERVIEW	GAYLE LAAKMANN MCDOWELL	CAREERCUP

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	Competitive Programming 4 – Steven Halim ,
OR-2	Introduction to Algorithms (CLRS) ,

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://codeforces.com/	Weekly contests, problem archive, editorials, rating system.
RW-2	https://leetcode.com/	Company-wise interview questions, online judge, contest platform.

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RW-3	https://www.codechef.com/	Monthly contests, DSA practice, beginner-friendly learning paths.
RW-4	https://cses.fi/problemset	Structured topic-wise practice for competitive programming.
RW-5	https://www.hackerrank.com/	Interactive coding challenges and skill assessments.
RW-6	https://usaco.guide/	Structured learning path for algorithms and competitive programming
RW-7	(https://cp-algorithms.com/)	Advanced algorithms with mathematical proofs and implementations

Software/Equipments/Databases		
Sr No	(S/E/D) (only if relevant to the course)	Salient Features
SW-1	https://github.com/	Store code templates, solutions, and collaborate using version control.
Virtual Labs (VL)		
Sr No	(VL) (only if relevant to the course)	Salient Features
VL-1	https://visualgo.org/	Interactive visualization of algorithms and data structures.

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Behaviour Analysis (Introduction to limits and behaviour of logic)	T-1 R-1	OR-2	Asymptotic Analysis (Big-O notation)	Student will be able to analyse complexity of any algorithm in terms of time and space.	Lecture cum demonstration, brain storming	Code with conditional statements and loops.
	Lecture 2	Behaviour Analysis (understanding taxonomy in worst case)	T-1 R-1	OR-1	Analysis of Algorithm	Student will be able to analyse complexity of any algorithm in terms of time and space	Discussion based on placement question with real time examples	Code with conditional statements and loops.

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Week 2	Lecture 3	Behaviour Analysis (analysing the effectiveness and efficiency of algorithms)	T-1 R-1	OR-2	To analyze the time and memory complexity	Student will be able to analyze the time and memory complexity of an algorithm or a structure	Lecture cum demonstration, brain storming	Code with conditional statements and loops.
	Lecture 4	Behaviour Analysis (measuring time and space complexity of algorithm)	T-1 R-1		complexity of algorithm	Student will be able to analyze the time and memory complexity of an algorithm or a structure	Discussion based on placement question with real time examples	Code with conditional statements and loops.
		Behaviour Analysis(trade-off concept)	T-1 R-1		complexity of algorithm	Student will be able to analyze the time and memory complexity of an algorithm or a structure	Live Demonstration through Example	Code with conditional statements and loops.
Week 3	Lecture 5	Primality Testing (Introduction to Primality Testing)	T-1 R-1	RW-7	Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms.	Scenario based problems discussion	Code with conditional statements and loops.
		Primality Testing($O(\sqrt{n})$) Algorithm for Primality Testing)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms.	Lecture, Live demonstrations with suitable example	Code with conditional statements and loops.
		Primality Testing (Factorization of a number)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms.	Scenario based problems discussion	Code with conditional statements and loops.
	Lecture 6	Primality Testing(Finding prime factors by taking the square root)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms	Scenario based problems discussion	Code with conditional statements and loops.
		Primality Testing(Binary Exponentiation)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms	Lecture, Live demonstrations with suitable example	Code with conditional statements and loops.
		Primality Testing(Fermat method)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms	Scenario based problems discussion	Code with conditional statements and loops.

Week 4	Lecture 7	Primality Testing(Sieve of Eratosthenes)	T-1 R-1		Different primality testing algorithms need to be discussed	Student will be able to test primality efficiently using different algorithms.	Scenario based problems discussion	
		Primality Testing (Segmented Sieve)	T-1 R-1		Different primality testing algorithms need to be discussed	Student will be able to test primality efficiently using different algorithms.	Scenario based problems discussion	
		Primality Testing(Mansi and her series)	T-1 R-1		Different primality testing algorithms need to be discussed	Student will be able to test primality efficiently using different algorithms.	Scenario based problems discussion	
	Lecture 8	Primality Testing (Collections of Pens)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms	Discussion based on real time examples	
		Primality Testing(next prime palindrome)	T-1 R-1		Different primality testing algorithms need to be discussed.	Student will be able to test primality efficiently using different algorithms	Scenario based problems discussion	
Week 5	Lecture 9	Recursion and Advanced Techniques(Introduction to recursion)	T-1 R-1	RW-1 RW-2 RW-3 RW-4	Basic Recursion (Difference between direct and indirect recursion)	Introduction to recursion and it's different types.	Discussion based on placement question with real time examples	Live demonstration with scenario based problem
		Recursion and Advanced Techniques(base condition)	T-1 R-1		Basic Recursion (Difference between direct and indirect recursion)	Introduction to recursion and it's different types.	Discussion based on placement question with real time examples	Live demonstration with scenario based problem
		Recursion and Advanced Techniques(Solving problems using recursion)	T-1 R-1		Basic Recursion (Difference between direct and indirect recursion)	Introduction to recursion and it's different types.	Discussion based on placement question with real time examples	Live demonstration with scenario based problem
		Recursion and Advanced Techniques(Classic and Modern Approaches)	T-1 R-1		Basic Recursion (Difference between direct and indirect recursion)	Introduction to recursion and it's different types.	Discussion based on placement question with real time examples	Live demonstration with scenario based problem
	Lecture 10	Recursion and Advanced Techniques(Direct vs. Indirect Recursion)	T-1 R-1		Introduction to recursion and it's different types.	Student will be able to apply the technique of recursion in different applications	Live demonstration with scenario based problem	

Week 5	Lecture 10	Recursion and Advanced Techniques(Tailed vs. Non-Tailed Recursion)	T-1 R-1		Introduction to recursion and it's different types.	Student will be able to apply the technique of recursion in different applications	Live demonstration with scenario based problem	
		Recursion and Advanced Techniques(Memory Allocation in Recursion)	T-1 R-1	RW-2 VL-1	Introduction to recursion and it's different types.	Student will be able to apply the technique of recursion in different applications	Live demonstration with scenario based problem	
		Recursion and Advanced Techniques(Advantages & disadvantages of recursive programming)	T-1 R-1		Introduction to recursion and it's different types.	Student will be able to apply the technique of recursion in different applications	Live demonstration with scenario based problem	
Week 6	Lecture 11				Test - Code based 1			
	Lecture 12	Recursion and Advanced Techniques(Backtracking)	T-1 R-1	RW-1 RW-2 RW-3 RW-4 SW-1	Introduction to recursion and it's different types	Student will be able to apply the technique of recursion in different applications.	Discuss the problem statement and provide a live demonstration using LeetCode	Solving a Sudoku puzzle. Finding path through maze. Placing queens on chessboard
		Recursion and Advanced Techniques(Permutations)	T-1 R-1		Introduction to recursion and it's different types	Student will be able to apply the technique of recursion in different applications.	Discuss the problem statement and provide a live demonstration using LeetCode	Solving a Sudoku puzzle. Finding path through maze. Placing queens on chessboard
		Recursion and Advanced Techniques(Combination Sum)	T-1 R-1		Introduction to recursion and it's different types	Student will be able to apply the technique of recursion in different applications.	Discuss the problem statement and provide a live demonstration using LeetCode	Solving a Sudoku puzzle. Finding path through maze. Placing queens on chessboard
		Recursion and Advanced Techniques(N-Queens)	T-1 R-1		Introduction to recursion and it's different types	Student will be able to apply the technique of recursion in different applications.	Discuss the problem statement and provide a live demonstration using LeetCode	Solving a Sudoku puzzle. Finding path through maze. Placing queens on chessboard
Week 7	Lecture 13	Recursion and Advanced Techniques(recursive problems- next happy number)	T-1 R-1		Introduction to recursion	Student will be able to apply the technique of recursion in different applica	Live demonstration with scenario based problem	Implementation of Recursive mechanism

Week 7	Lecture 13	Recursion and Advanced Techniques(sum string)	T-1 R-1		Introduction to recursion	Student will be able to apply the technique of recursion in different applica	Live demonstration with scenario based problem	Implementation of Recursive mechanism
		Recursion and Advanced Techniques(water overflow)	T-1 R-1		Introduction to recursion	Student will be able to apply the technique of recursion in different applica	Live demonstration with scenario based problem	Implementation of Recursive mechanism
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						
Week 8	Lecture 15	Basic Dynamic Programming(Introduction to Dynamic Programming)	T-1 R-1	RW-4 RW-5 RW-6	Introduction to Dynamic programming and its process	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	
		Basic Dynamic Programming(Fibonacci Sequence)	T-1 R-1		Introduction to Dynamic programming and its process	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programing
		Basic Dynamic Programming(Tiling problem)	T-1 R-1		Introduction to Dynamic programming and its process	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programing
		Basic Dynamic Programming(Climbing Stairs)	T-1 R-1		Introduction to Dynamic programming and its process	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programing

Week 8	Lecture 16	Basic Dynamic Programming(Tabulation vs Memoization)	T-1 R-1		Tabulation and Memoization, Tiling problem to discuss these techniques	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programming
		Basic Dynamic Programming(State Definition and State Transition)	T-1 R-1		Tabulation and Memoization, Tiling problem to discuss these techniques	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programming
		Basic Dynamic Programming(Optimal Substructure Property)	T-1 R-1		Tabulation and Memoization, Tiling problem to discuss these techniques	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live demonstration with scenario based problem	Implementation of Dynamic programming
Week 9	Lecture 17	Basic Dynamic Programming(Overlapping Subproblems Property)	T-1 R-1		How to solve dynamic problems with example	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Live demonstration with scenario based problem	Implementation of Dynamic programming
		Basic Dynamic Programming(Dynamic Programming Process and Techniques)	T-1 R-1		How to solve dynamic problems with example	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Live demonstration with scenario based problem	Implementation of Dynamic programming
		Basic Dynamic Programming(Formulating Dynamic Programming Problems)	T-1 R-1		How to solve dynamic problems with example	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Live demonstration with scenario based problem	Implementation of Dynamic programming
	Lecture 18	Dynamic Programming Problems(Longest increasing subsequence)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Live Demonstration through Example	
		Dynamic Programming Problems(Longest common subsequence)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Live Demonstration through Example	

Week 10	Lecture 19				Test - Code based 2			
	Lecture 20	Dynamic Programming Problems(Binomial coefficient)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Discussion based on placement question with real time examples	
		Dynamic Programming Problems(Box Stacking)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems	Discussion based on placement question with real time examples	
Week 11	Lecture 21	Dynamic Programming Problems(Integer Knapsack Problem (Duplicate Items Forbidden))	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stack	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Discussion based on placement question with real time examples	Live Demonstration through Example
		Dynamic Programming Problems(Edit Distance)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stack	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Discussion based on placement question with real time examples	Live Demonstration through Example
	Lecture 22	Dynamic Programming Problems(Matrix Chain Multiplication)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc.	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live Demonstration through Example	
		Dynamic Programming Problems(Balanced Partition)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc.	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live Demonstration through Example	
		Dynamic Programming Problems(Longest Increasing Subsequence (LIS))	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc.	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live Demonstration through Example	

Week 12	Lecture 23	Dynamic Programming Problems(Longest Common Subsequence (LCS))	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live Demonstration through Example	
		Dynamic Programming Problems(Balanced Partition Problem)	T-1 R-1		Dynamic Programming problems like Longest increasing subsequence, LCS, Box Stacking etc	Student will be able to apply Tabulation and Memoization in standard Dynamic Programming problems.	Live Demonstration through Example	
	Lecture 24	Efficient Sorting Algorithms & Analysis(Introduction to $O(n \log n)$ Sorting Algorithms)	T-1 R-1		Merge Sort iterative solution	use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	Merge sort
		Efficient Sorting Algorithms & Analysis(Iterative & Recursive Merge Sort)	T-1 R-1		Merge Sort iterative solution	use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	Merge sort
		Efficient Sorting Algorithms & Analysis(Quick Sort)	T-1 R-1		Quick sort	use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	Quick sort
Week 13	Lecture 25	Efficient Sorting Algorithms & Analysis(Sorting Elements by Frequency)	T-1 R-1		Counting sort, sorting array in strings sorting etc.	Student will be able to use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	
		Efficient Sorting Algorithms & Analysis(Finding Minimum Length Sorted Sub-array to Sort an Array)	T-1 R-1		Counting sort, sorting array in strings sorting etc.	Student will be able to use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	
		Efficient Sorting Algorithms & Analysis(Sorting Strings)	T-1 R-1		Counting sort, sorting array in strings sorting etc.	Student will be able to use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	
	Lecture 26	Efficient Sorting Algorithms & Analysis(case-specific sorting of strings)	T-1 R-1		Counting sort, sorting array in strings sorting etc.	Student will be able to use $n \log n$ sorting techniques and recursion in effective way	Live Demonstration through Example	

Week 13	Lecture 26	Efficient Sorting Algorithms & Analysis(Count Distinct Pairs with Difference of K)	T-1 R-1		Counting sort, sorting array in strings sorting etc.	Student will be able to use nlogn sorting techniques and recursion in effective way	Live Demonstration through Example	
Week 14	Lecture 27				Test - Code based 3			
SPILL OVER								
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:A0203 (2 best out of 3)

Component	Weightage (%)	Mapped CO(s)
Test - Code based 3	50	CO5, CO6
Test - Code based 2	50	CO3, CO4
Test - Code based 1	50	CO1, CO2

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Test - Code based 1	Unit1, Unit 2 syllabus will be considered for evaluation.	To evaluate the understanding of students related to Algorithm Analysis and Primality Testing.	Individual	Online	30	5 / 6
Test - Code based 2	To evaluate the understanding of students related to Recursion and Dynamic Programming.	Unit 3 and unit 4 syllabus will be considered for evaluation.	Individual	Online	30	9 / 10

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Test - Code based 3	To evaluate the understanding of students related to Advanced Dynamic Programming and Efficient Sorting Algorithms	Unit 5 and Unit 6 will be considered for the evaluation	Individual	Online	30	13 / 14
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List of suggested topics for term paper[at least 15] (Student to spend about 15 hrs on any one specified term paper)

Sr. No.	Topic
1	NA

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
Practical 1	List of Practical's	Finding prime factors by taking the square root	OR-2	Student will be able to test primality efficiently using different algorithms.
Practical 2	List of Practical's	Fermat method	OR-1 OR-2	Student will be able to test primality efficiently using different algorithms.
	List of Practical's	Sieve of Eratosthenes	RW-1 RW-2 RW-3 RW-4 RW-5	Student will be able to test primality efficiently using different algorithms.
Practical 3	List of Practical's	Segmented Sieve	RW-2	Student will be able to test primality efficiently using different algorithms.
	List of Practical's	Tiling problem	RW-2	Student will be able to test primality efficiently using different algorithms.
Practical 4	List of Practical's	Longest increasing subsequence	RW-1 RW-4	apply Tabulation and Memoization in standard Dynamic Programming problems.
	List of Practical's	Longest common subsequence	RW-2 RW-7	apply Tabulation and Memoization in standard Dynamic Programming problems.
Practical 5	List of Practical's	Binomial coefficient	RW-6 RW-7	apply Tabulation and Memoization in standard Dynamic Programming problems.

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Practical 6	List of Practical's	Box Stacking	RW-2 RW-3 SW-1 VL-1	apply Tabulation and Memoization in standard Dynamic Programming problems.
Practical 7	List of Practical's	Integer Knapsack Problem (Duplicate Items Forbidden)	OR-2 RW-2 VL-1	apply Tabulation and Memoization in standard Dynamic Programming problems.
Practical 8	List of Practical's	Edit Distance	RW-2	apply Tabulation and Memoization in standard Dynamic Programming problems.
Practical 9	List of Practical's	Matrix Chain Multiplication	RW-2	apply Tabulation and Memoization in standard Dynamic Programming problems
Practical 10	List of Practical's	Balanced Partition	RW-1 RW-5	apply Tabulation and Memoization in standard Dynamic Programming problems.
Practical 11	List of Practical's	Merge Sort	RW-1 RW-2 RW-6	Student will be able to use $n \log n$ sorting techniques and recursion in effective way
Practical 12	List of Practical's	Quick Sort	RW-2	Student will be able to use $n \log n$ sorting techniques and recursion in effective way
Practical 13	List of Practical's	Counting Sort	RW-1 RW-6	Student will be able to use $n \log n$ sorting techniques and recursion in effective way
Practical 14	List of Practical's	String sorting	RW-1 RW-2 RW-3	Student will be able to use $n \log n$ sorting techniques and recursion in effective way
	SPILL OVER			
Practical 15	Spill Over			